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6888/02

October/November 2025

1 hour 15 minutes

Additional materials: Calculator

Write your name, centre number and candidate number in spaces provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs, tables or rough working.
Do **not** use staples, paper clips, highlighters, glue or correction fluid.
Do **not** write on the bar code.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

<i>For Examiner's Use</i>	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total	

[Turn over

1 Fig. 1.1 shows part of five structures, **A**, **B**, **C**, **D** and **E**.

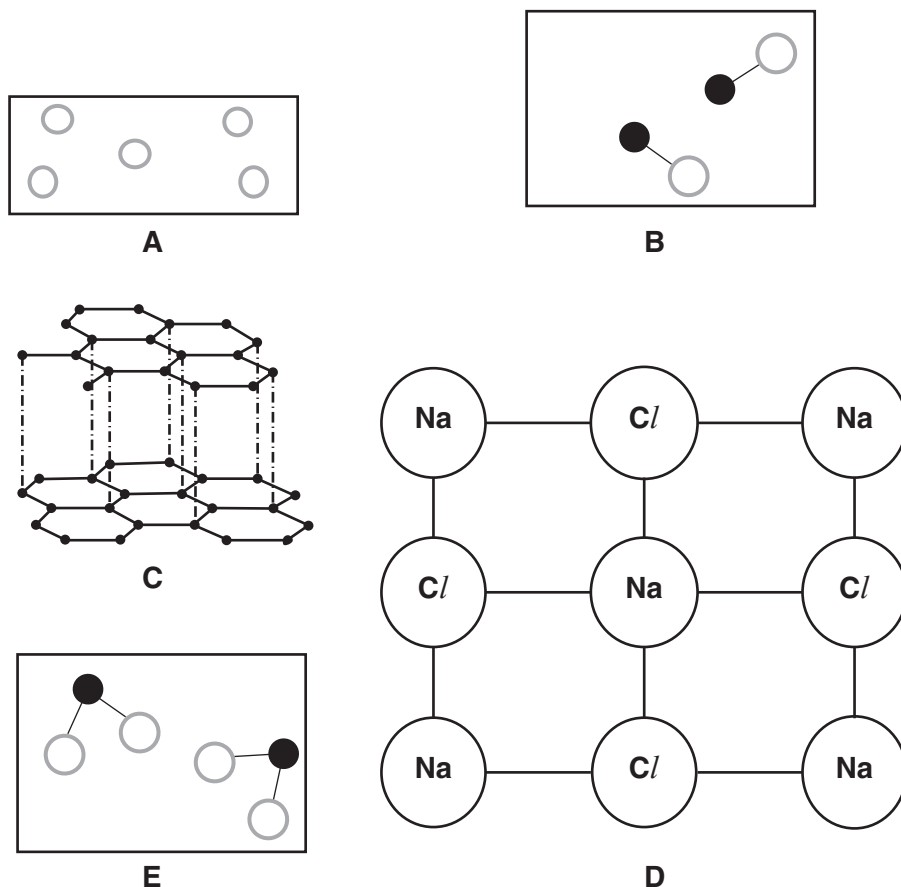


Fig. 1.1

Identify the structure, **A**, **B**, **C**, **D** or **E**, that:

- (a) conducts electricity when solid [1]
- (b) is used for filling weather balloons [1]
- (c) is a solid at room temperature that dissolves in water [1]
- (d) is one of the reactants during photosynthesis [1]

- 2 (a) Fig. 2.1 shows a girl cycling along a level road at a constant velocity.

The forward driving force is F_D and the total backward force is F_B .



Fig. 2.1

- (i) Speed is a scalar quantity and velocity is a vector quantity.

State what is meant by a *vector quantity*.

.....
 [2]

- (ii) Explain, in terms of the forward driving force and the total backward force, why the girl is travelling at a constant velocity.

.....

 [2]

- (iii) Name the forces that cause the total backward force, F_B .

1
 2
 [2]

- (b) Fig. 2.2 shows the velocity-time graph for the girl travelling between two stations G and H.

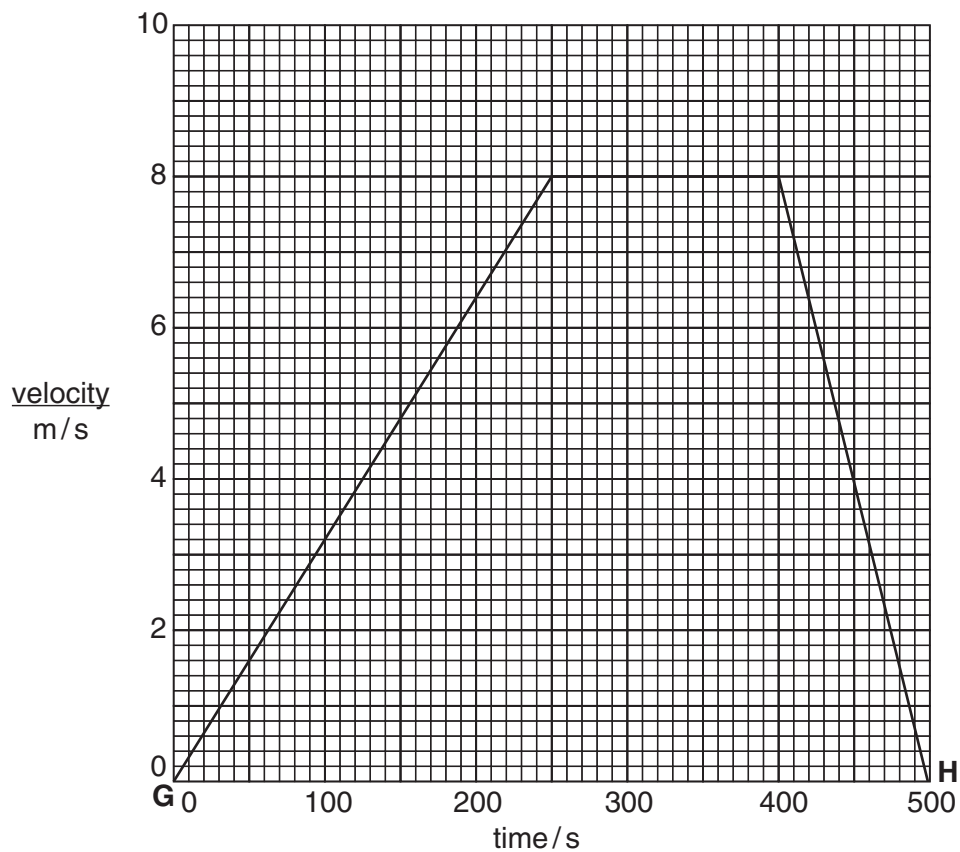


Fig. 2.2

- (i) Using Fig. 2.2, describe the motion of the girl during the first 400 seconds.

.....

 [2]

- (ii) Determine the distance covered by the girl from 400 seconds to 500 seconds.

distance m [2]

3 Fig. 3.1 shows the chromatogram of spinach juice.

The spinach juice separates into four dyes known as carotene, xanthophyll, chlorophyll-a and chlorophyll-b.

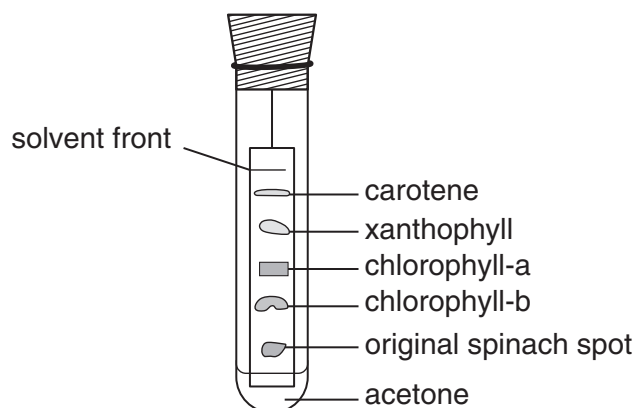


Fig. 3.1

(a) Name the most soluble spinach dye in Fig. 3.1.

..... [1]

(b) The solvent front (distance moved by solvent) is 12 cm from the original spot.

Calculate the retardation factor, R_f , of xanthophyll if it travels 8 cm from the original spot.

Use the formula:

$$R_f = \frac{\text{distance moved by solute}}{\text{distance moved by solvent}}$$

..... [2]

(c) Another component of spinach, lutein, is colourless and least soluble.

(i) Describe how lutein can be identified on the chromatogram.

..... [1]

(ii) On Fig. 3.1, mark the correct position of lutein on the chromatogram, using the letter J.

[1]

(d) Explain why the spinach juice is **not** a pure substance.

.....
..... [1]

- 4 (a) Fig. 4.1 shows a ray of light, incident ray, on a parallel sided glass block.

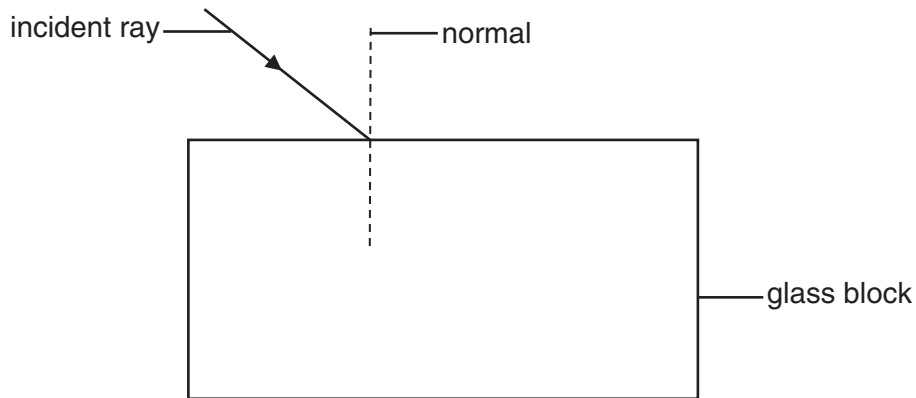


Fig. 4.1

- (i) Mark the angle of incidence, i , in Fig. 4.1. [1]
- (ii) On Fig. 4.1, complete the path of the incident ray as it enters and leaves the glass block. [2]
- (b) A chart showing all the electromagnetic waves in order of ascending or descending wavelengths is called the electromagnetic spectrum.

Fig. 4.2 shows different waves in the electromagnetic spectrum.

radio waves	microwaves	K	visible light	ultra-violet	L	gamma rays
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Fig. 4.2

- (i) Name the types of waves represented by **K** and **L** in Fig. 4.2.
- K**
- L** [2]
- (ii) From Fig. 4.2, identify the wave with the shortest wavelength.
- [1]
- (iii) Electromagnetic waves travel through a vacuum.
- State **one** other property common to all electromagnetic waves.
-
- [1]

- (c) Sound is also a wave. It cannot be transmitted in a vacuum.

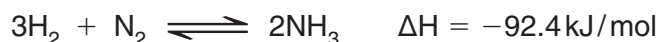
Explain why sound cannot be transmitted in a vacuum.

.....

.....

..... [2]

- 5 Ammonia is formed by reacting hydrogen and nitrogen gas according to the equation:



- (a) Name the process used in the manufacture of ammonia.

..... [1]

- (b) A catalyst is used to increase the rate of the reaction between hydrogen and nitrogen.

- (i) Name the catalyst used in the manufacture of ammonia.

..... [1]

- (ii) Using collision theory, explain how the catalyst increases the rate of the reaction.

.....

.....

..... [2]

- (c) Ammonia is ideally manufactured at a temperature of 450 °C.

Explain the effect of increasing the temperature to 600 °C on the yield of the ammonia.

effect on yield

explanation

.....

..... [3]

- 6 Fig. 6.1 shows a conductor placed between the poles of magnets and connected to a galvanometer, **G**.

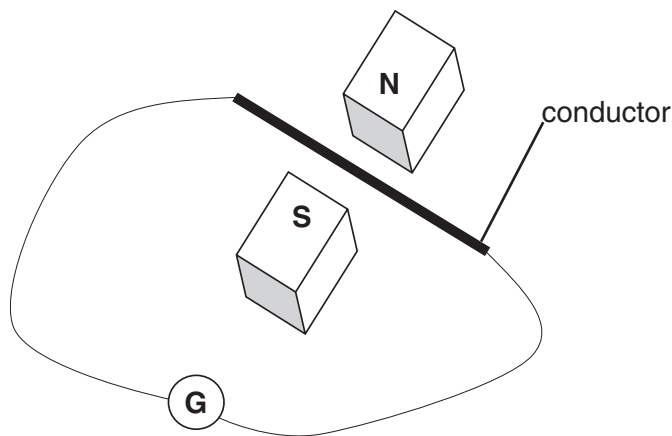


Fig. 6.1

- (a) Name the principle demonstrated by moving the conductor between the poles in Fig. 6.1.

..... [1]

- (b) The conductor is moved vertically upwards between the poles of the magnets in Fig. 6.1.

State and explain the effect of this movement on the galvanometer.

.....

 [3]

- (c) The conductor is moved vertically upwards between the poles of the magnets in Fig. 6.1 at a higher speed.

State the effect of this movement on the galvanometer.

.....
 [1]

- (d) The conductor is stationary between the poles of the magnets in Fig 6.1.

State and explain the effect on the galvanometer.

.....

 [2]

- (e) Name a device that uses the same principle as the one shown in Fig. 6.1.

..... [1]

- 7 Lithium reacts with nitrogen to form lithium nitride, Li_3N .

- (a) Write the electronic configuration of a lithium ion.

..... [1]

- (b) Explain why lithium atoms are very reactive.

.....

.....

..... [2]

- (c) Draw a 'dot and cross' diagram to show the bonding in Li_3N .

Show outer electrons only.

[2]

- (d) Nitrogen reacts with oxygen to form oxides of nitrogen.

1.4 g of nitrogen reacts with 3.2 g of oxygen to form an oxide of nitrogen.

- (i) Calculate the empirical formula of the oxide of nitrogen, N_xO_y .

..... [3]

- (ii) The oxide of nitrogen has a molecular mass of 92.

Determine the formula of this oxide of nitrogen.

..... [2]

- 8 Electric heaters are commonly used as a source of heat.

- (a) Describe how heat is produced in the element of an electric heater.

.....

 [2]

- (b) An electric heater is rated 240V, 1200W.

- (i) Calculate the current in the electric heater when working normally.

State the unit.

..... [3]

- (ii) The heater contains a fuse to protect both the appliance and the user when a fault develops.

Describe how a fuse protects the appliance and the user.

.....

 [2]

- (c) Some fluorescent lamps use electricity.

State **one** hazard of broken fluorescent lamps.

.....
 [1]

- 9** Aluminium appears to be unreactive due to its oxide layer of aluminium oxide. Aluminium oxide behaves both as an acid and a base.

(a) State the term used to describe oxides that behave both as an acid and a base.

..... [1]

(b) Name a reagent that reacts with aluminium oxide when it behaves as a base.

..... [1]

(c) In the box below, draw a diagram to show the metallic bonding in aluminium.



[3]

10 Fig. 10.1 shows a combination of logic gates for a combination latch lock.

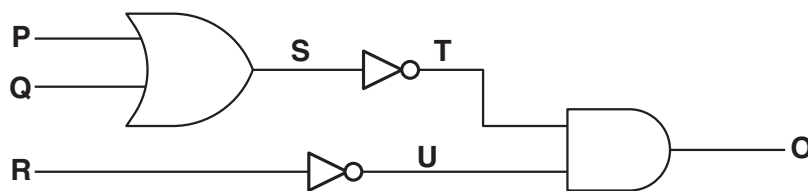


Fig. 10.1

(a) Circle an **OR** logic gate in Fig. 10.1. [1]

(b) (i) Table 10.1 shows a truth table for some combinations of the latch lock.

Table 10.1

P	Q	R	S	T	U	Output O
1	1	1				
1	0	0				
0	0	0				

Complete Table 10.1 for the combinations of **P**, **Q** and **R**. [3]

(ii) Identify the input combination of **P**, **Q** and **R** that opens the lock.

..... [1]

11 Paraffin is a hydrocarbon that is obtained from petroleum.

(a) Name the process used to extract paraffin from petroleum.

..... [1]

(b) Paraffin can be used to form short-chain hydrocarbons such as ethene.

Describe the conditions necessary to convert paraffin into ethene.

.....

 [2]

(c) Ethene is a member of a homologous series.

(i) Name the homologous series to which ethene belongs.

..... [1]

(ii) Draw the structure of the fourth member of this homologous series.

[2]

(d) Fats are natural polymers made by condensation polymerisation between glycerol and 3 fatty acids.

Draw the structure of the fat formed between glycerol and the 3 fatty acids shown in Fig. 11.1.

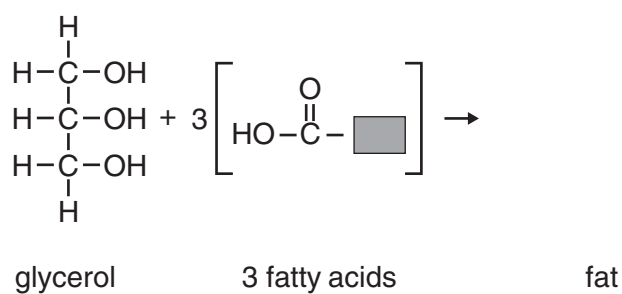


Fig. 11.1

[2]

DATA SHEET
The Periodic Table of the Elements

Group																			
I	II											III	IV	V	VI	VII	0		
		1 H Hydrogen 1																	4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	101 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	209 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	

* 58–71 Lanthanoid series
† 90–103 Actinoid series

a

X

a = relative atomic mass
X = atomic symbol
b = atomic (proton) number

140	141	144	147	150	152	157	159	163	165	167	169	173	175
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium
58	59	60	61	62	63	64	65	66	67	68	69	70	71
232	231	238	237	244	243	247	247	251	252	257	258	259	260
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
Thorium	Protactinium	Uranium	Neptunium	Plutonium	Americium	Curium	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium
90	91	92	93	94	95	96	97	98	99	100	101	102	103

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

